Life Cycle Assessment of HPCG on the IRIS and Aion Clusters: Environmental Impact and Optimisation

Fabio Dollaku
Master in High Performance Computing
University of Luxembourg
fabio.dollaku.001@student.uni.lu

Abstract—This paper assesses the carbon footprint of the HPCG benchmark on the two clusters of the University of Luxembourg, air-cooled IRIS and liquid-cooled Aion, and then tries to lower it without changing the benchmark's code. Following ISO 14040 and 14044, HPCG was run 100 times on one reserved node of each cluster with 28 processes. The energy of every run was read from the processors' counters, and the site's overhead and the node's manufacture were added. On average a run emits 6.0 gCO_2e on IRIS and 3.1 g on Aion. Embodied and operational emissions stand at 1.0 to 1 on IRIS and 1.4 to 1 on Aion, so the time a run holds its node matters as much as the power it draws. With the assumed facility efficiencies, site overhead accounts for 36 % of the estimated gap between the clusters. HPCG's own build already compiles with `-O3`. Of the settings a user controls, compiler flags, a cap on the processor clock and the placement of the processes, the 2.6 GHz clock cap cuts a run by 8.7 % on IRIS and one process per L3 cache with the 2.0 GHz cap cuts a run by 20.0 % on Aion, while moving the job from IRIS to Aion cuts it by 48 %.

I. Introduction

Users of a shared cluster rarely see what their jobs cost the environment. The scheduler reports time, sometimes energy, but never emissions. The University of Luxembourg runs two clusters side by side: IRIS, in service since 2017, with Intel processors cooled by air, and Aion, in service since 2021, with AMD processors cooled by liquid. A user who sends a job to one of them picks a processor, a memory system and a cooling technology at once, usually without noticing.

The job studied here is HPCG, the High Performance Conjugate Gradients benchmark [1]. It solves the kind of large sparse linear system found in many simulation codes, a set of equations in which most coefficients are zero. Its speed depends more on the memory than on the processor, which makes it a good test of how a whole node, not just its processor, turns electricity into computation. The code is HPCG's public reference implementation, https://github.com/hpcg-benchmark/hpcg, used unmodified at commit `114602d458d1034faa52b71e4c15aba9b3a17698`, with the fixed input of Table I. The method is life cycle assessment, standardised in ISO 14040 [2] and 14044 [3]. Besides the energy the node draws during the run, it counts the site's cooling and power distribution, and the run's share of the node's manufacture.

The research question is: how large is the mean carbon footprint of a run of HPCG on IRIS and on Aion, how much of the difference between the two clusters comes from the hardware and how much from the cooling, and how much of the footprint can the settings a cluster user controls remove without touching the code? The submission scripts, job logs and results are available in the project repository at https://github.com/NexusSin/hpcg-ulhpc-carbon-footprint.git.

II. Related Work

This study draws on four strands of work.

Estimating the footprint of computation. Lannelongue et al. [4] reduce the footprint of a computation to quantities a user can supply without measuring anything, such as run time, the cores and memory used, a usage factor, a power draw per core and per gigabyte, a PUE and a grid intensity. By their own account the estimate counts the electricity of the task and leaves out the manufacture of the hardware. Here the energy is measured on the node and the manufacture is added; on IRIS and Aion it turns out to be half of the footprint or more (Section III-E).

Reading energy from the processor. The inventory rests on the processors' built-in energy counters. Khan et al. [5] compared Intel's counters with power measured at the plug and found the two highly correlated, at a negligible cost to the running code. They tested Intel processors only; how accurate the counters are on Aion's AMD processors is still open, and Section VI counts it among the threats.

What HPCG measures. Dongarra, Heroux and Luszczek [6] introduced HPCG because LINPACK, the benchmark behind the TOP500 list, no longer reflects the simulations that drive large machines. Those simulations solve sparse systems, which stress the memory more than the peak arithmetic of the processor. HPCG reports a rate of floating-point operations and says nothing about the energy behind it, which is what this paper adds.

Life-cycle models of computing. Wadenstein and Vanderbauwhede [7] combine the embodied carbon of servers, the electricity they draw and the site's PUE in one life-cycle model. Where electricity is low in carbon, they find, the manufacture of the hardware dominates. Their model covers a computing centre over decades; the same three terms are applied here to a single job.

III. Life Cycle Assessment

The five subsections follow the four phases of ISO 14040 [2] and the reporting that ISO 14044 [3] requires after them. All tables are in Appendix A.

A. Goal and scope definition

The goal is to quantify the mean environmental footprint of a run of HPCG, over 100 runs on each of IRIS and Aion, two clusters that differ in processor architecture, memory, node design and cooling technology, and to identify the sources of the difference. Running the same job on both lets us compare the measured node energy and estimate the contributions of facility overhead and hardware manufacture under the stated assumptions. The study is aimed at users choosing where to run jobs and at system operators.

Functional unit. One run of HPCG with a fixed input, submitted through SLURM, the clusters' job scheduler, on one compute node reserved for the job alone, with the same number of parallel processes (MPI ranks) on both clusters. Since HPCG adapts its amount of work to a fixed duration, results are reported per run and per unit of computational work. The job is run 100 times on one node of each cluster for the baseline. The means and standard deviations describe the difference between the two nodes and the variation within each set of runs.

Impact criterion. The carbon footprint in CO_2-equivalent (IPCC 2021 [8], GWP100). Energy, water and hardware use are also reported in physical units.

System boundary. Every part of the cluster that the job uses, in four components: the compute node, over its whole life cycle; the interconnect and the shared storage, in their use phase; and the site, comprising power distribution, cooling and the shared services, such as the login node and the scheduler, through which the job passes. The user's workstation and the one-off compilation of the code are excluded.

B. Life cycle inventory

The inventory quantifies the physical flows of a run, per component and as the mean of the 100 runs. It rests on primary data measured on the hardware during the runs, with secondary data only where a quantity cannot be measured from within a job.

Measurement protocol. On each cluster the job runs 100 times back to back in one allocation, on a node reserved for it alone so that no other workload enters the readings. No run is discarded. With a hundred runs the mean is steady enough for the difference between the clusters to stand out from the variation between runs. The footprint is computed run by run and reported as the mean of the 100 runs, with its standard deviation. Test jobs in September 2026 established which energy readings each cluster offers: IRIS's processors report the energy of their packages and of the memory, Aion's of their packages only. Section VI sizes the difference this makes.

Allocation. Burdens shared by many jobs, such as a node's manufacture, are allocated to a run in proportion to the time it holds the node, out of all the time the node is allocated to jobs over its service life.

1) Compute node: Processor energy, and memory energy where the hardware reports it, is read from the processors' built-in energy counters (RAPL) before and after each run, with the scheduler's energy record as a cross-check where available. During each run the system also records the run time, processor utilisation and clock frequency, chip and network-card temperatures, the memory reserved and used, and local disk traffic; idle power is measured before and after the runs, and HPCG reports the work done. The power supply and fans cannot be metered from within a job, a gap stated with the results. Embodied emissions come from the manufacturer's product carbon footprint where one exists and from a modelled substitute otherwise, both secondary sources.

2) Interconnect: Traffic through the node's network interfaces is read from their port counters in every run; the job, confined to one node, exchanges no data with other compute nodes. The port counters include background traffic. The measurements provide no per-job estimate of switch electricity.

3) Storage: The run directories preserve the benchmark output files. Their sizes measure the output saved on disk. The scheduler records bytes at the system-call level, which can include cached reads. These records do not measure physical traffic or electricity at the shared file servers.

4) Site: Cooling and power distribution cannot be metered from within a job. They enter through the power usage effectiveness (PUE), the ratio of facility to IT energy. Neither room publishes a measured PUE, so the values come from European averages for rooms of the same size and from the operator's design target; water use is estimated from the same averages.

C. Life cycle impact assessment

The impact assessment turns the flows of the inventory into an environmental impact. It uses a single criterion, climate change, characterised with the IPCC 2021 method [8] and expressed as the global warming potential over 100 years (GWP100), in grams of CO_2-equivalent. Every factor is read in this indicator; a source that counts carbon dioxide alone, follows an earlier IPCC report or does not say which is flagged where it is used. A carbon footprint leaves out the other categories of a full assessment, among them water use, the depletion of minerals and metals, and toxicity. Water and hardware use still appear in physical units next to the footprint, but they are not converted into an impact.

Inventory values. Table III summarises the 100 runs on each cluster. An average IRIS run holds its node for 217 s and draws 61.0 kJ (standard deviation 0.4 kJ) in the processor packages and the memory, 281 W on average. An Aion run takes 126 s and 40.1 kJ, or 317 W, all of it in the packages, since Aion's processors report no memory energy. Table V gives the mean of every other quantity measured per run.

Factors. Three factors carry the inventory to grams (Table II). For the grid I declare 108.42 gCO_2 per kWh, the intensity that the Luxembourg regulator, ILR, publishes for

all the electricity supplied to final customers in the country in 2025 [9], read on 23 October 2026. Luxembourg imports most of its electricity, so I prefer a figure for the electricity consumed in the country, imports included, to one for the electricity produced there. This one counts only the carbon dioxide of generation, a limit that Section VI discusses, and it applies to the operational term alone, since the hardware was made in other countries.

For the PUE I declare 1.63 for IRIS, the average that a 2025 assessment for the European Commission reports for data centres of 100 to 500 kW [10], the class of IRIS's 300 kW air-cooled room [11]. No measured value is published for Aion either. Its racks are cooled directly by liquid. In every blade a cold plate, fed by a separate hot-water circuit, takes the place of fans, and the operator describes the design as targeting a PUE very close to 1 [12], [11]. I declare 1.1 for Aion, an assumption that Section VI tests. Applying PUE to the measured processor and memory energy estimates the associated facility overhead, including cooling.

For manufacture, transport and end of life, an IRIS node carries 1 256 kg CO_2e: the 19.3 % of Dell's 6 510 kg product carbon footprint for the PowerEdge C6420 that lies outside its use phase [13]. The use phase is left out because it is measured. No such figure exists for the Aion node, so I substitute the 1 200 kg that Wadenstein and Vanderbauwhede model for a recent two-socket AMD server [7]. A node's allocated life is its years in service so far, 8.8 for IRIS's Skylake nodes (since December 2017) and 4.9 for Aion (since late 2021) [14], [12], times the share of each cluster allocated to jobs over the last twelve months, 32.94 % and 54.37 % (Table VI). This gives 2.90 and 2.66 years.

From joules to grams. The footprint of a run adds an operational and an embodied term:

$$C_{\text{run}} = \underbrace{E_{\text{IT}} \cdot \text{PUE} \cdot I_{\text{grid}}}_{\text{operational}} + \underbrace{C_{\text{emb}} \cdot \frac{t_{\text{run}}}{T_{\text{alloc}}}}_{\text{embodied}}$$

Here E_{IT} is the energy measured on the node during the run, converted from joules to kWh (1 kWh = 3.6 MJ), and I_{grid} is the grid factor in gCO_2e per kWh. In the embodied term, C_{emb} is the node's embodied footprint in gCO_2e, t_{run} the time the run holds the node, and T_{alloc} the time the node has been allocated to jobs over its service life so far, both in seconds. Dividing by the allocated time instead of the full service life charges the hours a node stands idle to the jobs that do run. The estimate omits the unmetered IT electricity of switches and shared storage.

For the mean IRIS run, E_{IT} is 61.0 kJ, or 16.9 Wh. At 108.42 g per kWh this is 1.8 g for the node's electricity, and a PUE of 1.63 adds 1.2 g for cooling and power distribution. Holding the node for 217 s of its 2.90 allocated years carries 3.0 g of its manufacture. Averaged over its 100 runs, a run of HPCG thus emits 6.0 gCO_2e on IRIS (standard deviation 0.03 g), and over the 100 runs on Aion 3.1 g (1.2 g of electricity, 0.12 g of site overhead and 1.8 g of manufacture; Table IV). These figures are given to two significant digits, the precision of the factors.

D. Interpretation

Interpretation, in the sense of ISO 14044, identifies the significant issues, checks the result for completeness, sensitivity and consistency, and draws conclusions against the goal of Section III-A. The figure of Section III-C is the mean carbon footprint of a run of HPCG with a fixed input, over 100 runs on one reserved node of each cluster, including the site's overhead and the run's share of the node's manufacture.

Embodied versus operational. The most significant issue is the ratio between the two terms of the formula. On IRIS a run's share of the node's manufacture, 3.0 g, equals its operational emissions, 3.0 g of electricity and site overhead together: a ratio of 1.0 to 1, or 50 % embodied against 50 % operational. On Aion the ratio is 1.4 to 1, 58 % against 42 %: its low PUE removes most of the site overhead, and manufacture becomes the larger part. The ratio is not a property of the job alone. The grid factor scales only the operational term, the node's allocated life only the embodied one, and the run time scales both. At the EU-27 average of 209 g per kWh [15] the ratio falls to 0.5 to 1 on IRIS and 0.7 to 1 on Aion, and operation dominates; on Luxembourg's low-carbon supply, manufacture weighs as much as electricity or more. A run's length therefore counts as much as its power, which decides where Section IV looks for a saving.

Meaning. At the declared grid factor, an hour of a 5 W LED bulb emits 0.54 g. An average run on IRIS then amounts to 11 hours of that bulb, 21 m in a small petrol car at 291 g per km, or 0.021 % of the 28 kg that a Luxembourg resident emits in a day [15]. An average run on Aion amounts to 5.8 hours, 11 m or 0.011 %.

Variation between runs. The 100 runs on each node barely differ. Between the lowest and the highest, their energy varies by 4.5 % of the mean on IRIS and by 6.3 % on Aion (Table III). On Aion, runs 16 to 46 took about 4 s longer at an unchanged clock and load. The records do not identify the cause of that slowdown. Those runs stay in the mean. The first run of each set was no slower than the rest, so a cold start does not show. The gap between the clusters is far larger than any of this, so it does not come from chance between runs.

Comparing the clusters. On average a run on IRIS emits 1.9 times as much as one on Aion, and per unit of work the gap is wider still. Per second of node time IRIS emits only 1.1 times as much, but Aion's rating is 2.5 times higher. Dividing the whole-run footprint by the floating-point work reported for the timed phase gives 3.8 times as much carbon per GFLOP on IRIS as on Aion (Table III). Of the estimated gap of 2.8 g per run, the model assigns 36 % to site overhead at the declared PUEs, 22 % to the node's electricity and 41 % to its manufacture. Memory is the likeliest reason why IRIS's run takes longer, which raises both its electricity and its share of manufacture, although two machines cannot separate memory from processors (Section VI). HPCG is limited by memory traffic, and an IRIS node fills only four of the six memory

channels of each processor, for a peak of 154 GB/s, where an Aion node fills all sixteen of its channels with faster memory, for 410 GB/s [14], [12] (Table I). Aion loses on another count: the job keeps 28 of its 128 cores busy, 21.8 % of the node's processing time, and the idle cores still draw power.

Checks. A top-down estimate tests the counter readings: the processors' rated power times the time a run holds the node. On IRIS that is 280 W times the run time, 60.8 kJ, and the packages drew 92 % of it, a plausible level for processors kept fully busy. On Aion it is 560 W times the run time, 70.8 kJ, of which the packages drew 57 %, as expected when most cores idle. Neither reading exceeds its rated value or hints at a fault, and the result keeps the counter readings. On IRIS the scheduler's own energy record agrees with the counters within 0.14 %. Because the scheduler reads the same registers, the agreement checks how they are read rather than the sensors themselves. Finally, the sensitivity check of Section VI recomputes the footprint with other grid factors and PUE values: the order of the two clusters holds in every case.

E. Reporting

Averaged over 100 runs on each cluster, a run of HPCG with the fixed input, on one node reserved for it with 28 processes, emits 6.0 gCO_2e on IRIS and 3.1 g on Aion (GWP100). The conventions behind these figures are a grid factor of 108.42 gCO_2 per kWh, a PUE of 1.63 for IRIS and 1.1 for Aion, and the node's manufacture charged by the time the run holds the node out of its allocated life. A reader who prefers other factors can recompute the footprint from the energy per run, 61.0 kJ and 40.1 kJ at the counters, with the formula of Section III-C, without repeating any measurement.

On both clusters the hotspot is the node's manufacture, with 50 % of the IRIS figure and 58 % of Aion's; embodied and operational emissions stand at 1.0 to 1 on IRIS and 1.4 to 1 on Aion. The node's electricity comes second, with 31 % and 39 %. Both grow with the time a run holds the node. At the assumed PUEs, site overhead accounts for 19 % of the total on IRIS and 4 % on Aion.

IV. Optimisation

A. Optimisation procedure

The hotspot of Section III-E is the node's manufacture, charged by the time a run holds the node, and the node's electricity comes second. A change can therefore lower the footprint in two ways: shorten the run, which lowers both terms, or lower the power drawn without lengthening the run by as much, which lowers the operational term only. HPCG's code and input stay as they are, and so do the 28 processes and the node held for the job alone; only what a cluster user controls changes: how the code is compiled, how fast the processors clock, and where on the node the processes run.

Compiler flags. HPCG's own build configuration, used for every run of Section III-C, already compiles with `-O3`, the highest of GCC's standard optimisation levels, together with `-ffast-math` and automatic vectorisation [16] (Table I). Adding `-O3` therefore changes nothing. To see what it contributes, the same build is measured at `-O2`. The one common flag the configuration leaves out is `-march=native`, which lets the compiler use every instruction of the processor that builds the code [17]. Both were tested on Aion.

Clock cap. HPCG spends much of its time waiting for memory (Section III-D), so a processor that clocks lower may lose little speed while drawing less power. SLURM lets a job cap the clock of its own cores with the `--cpu-freq` option of `srun` [18]. On Aion, whose processors offer only 2.6, 2.0 and 1.5 GHz steps, the caps tested are 2.0 and 1.5 GHz. On IRIS, whose cores otherwise run at about 3.3 GHz with their turbo boost (Table V), the cap is 2.6 GHz, the processors' base clock, which switches the boost off.

Placement. An Aion node has eight memory domains of 16 cores, and each domain four groups of four cores that share a 16 MB L3 cache. By default SLURM puts the job's three or four processes in each domain on the domain's first group, so they share one cache while three groups stay idle, as the job's record of each process's cores shows. The spread placement keeps every process in its domain but gives each its own group, and so its own cache. On IRIS the 28 processes fill the node's 28 cores, and placement has nothing to change.

Before and after. The before is the 100 runs of the unchanged job in Section III-C; the after is 100 runs of the changed job, with the same protocol, so that the two are measured the same way. On Aion, the change was chosen in short rounds on one node, each candidate alternating with the unchanged job in an order that rotated from round to round, so that a slow drift of the node favours none of them: 10 rounds for the compiler flags and clock caps and 2 for the placement, whose difference was clear from the first round. A change is kept only if HPCG validates every run, every run does at least the baseline's work, and the footprint is lower in every round. "At least": HPCG fixes its duration, not its work, so a configuration that runs faster may complete an extra set of iterations, which is extra work at no extra charge. On IRIS, the 2.6 GHz cap was chosen without such rounds. IRIS nodes differ widely, though: in this project's earlier, shorter measurements on 5 IRIS nodes, the same run emitted between 4.8 and 7.2 g, against 3.1 to 3.2 g on 4 Aion nodes. Section III-C's IRIS node was not free in time, so the job first ran the unchanged and the capped job alternately 10 times on the node it was given, and the 100 capped runs are compared with the unchanged job on that node. The lecturer suggested the compiler flags; I asked for the 100 runs after optimisation, for the comparison to rest on them, for a change that lowers the emissions even if the run gets slower, and for the IRIS cap without a test. The AI coding agent proposed the `-O2` control, the clock caps and the placement, wrote the build variants, the job script and the script that applies the rule, and submitted and analysed the jobs; I accepted the levers and the rule for keeping a change.

B. Implementation

Nothing in HPCG's source tree or in `hpcg.dat` changed. The compiler variants are passed to `make` on the command line, as `CXXFLAGS` with `-O2` in place of `-O3`, or with `-march=native` appended, and each is built from the same commit in its own folder at the start of the allocation. The clock cap is one option of the `srun` line that starts HPCG, `--cpu-freq=<lowest>-<cap>:Performance`, with the performance governor that the clusters allow; the job records the limit under which each run actually ran. The placement is another option of the same line, `--cpu-bind=map_cpu:<cores>`, with `--whole` so that the run may use any core of the node the job holds, and every run records the cores of its 28 processes. A script applies the rule to the rounds' footprints, computed with the code that produces every figure of this paper. Two options were rejected. HPCG's build switch for contiguous matrix storage, `-DHPCG_CONTIGUOUS_ARRAYS`, selects different code paths in the source, which the brief rules out. Changing the problem size, the run time or the number of processes would change the functional unit.

To check that the job still produces the same result, every run's report is read for HPCG's verdict, the number of conjugate-gradient iterations and the scaled residual. A configuration that HPCG does not validate, or that does less work, is no longer the same job.

C. Footprint after optimisation

Table VII compares the 100 runs before and after optimisation; Table VIII gives the rounds that chose the change. Footprints in this section carry three digits so that the changes show; as absolute figures they are accurate to two, the precision of the factors (Section III-C). On IRIS, HPCG validated all 20 runs of its rounds, all with the same scaled residual. Every configuration did the baseline's 100 iterations. On Aion (flags and clock), HPCG validated all 50 screening runs, all with the same scaled residual. Every configuration did at least the baseline's 200 iterations except the 1.5 GHz cap, which did 150 in 5 of its 10 runs. HPCG repeats its set of 50 iterations as many times as fit in the 60 s that `hpcg.dat` requests, plus one, so a configuration that runs a set slightly slower can do one set fewer, which the rule of Section IV-A excludes, and a faster one a set more. On Aion (placement), HPCG validated all 8 screening runs, all with the same scaled residual. Every configuration did the baseline's 200 iterations.

Compiler flags. Going back from `-O3` to `-O2` changed the footprint of a run by +0.1 % on Aion; adding `-march=native` changed it by −0.1 % on Aion. On Aion the two levels cannot be told apart from the variation between rounds, and `-march=native` made no difference beyond that variation.

Clock cap. At 2.0 GHz on Aion, the power changed by −16.7 % and the run time by +3.4 %, so the energy changed by −13.9 %, the operational emissions by −13.9 %, the manufacture charged to the run by +3.4 %, and the footprint by −3.9 % (lower in every round). At 1.5 GHz on Aion, 5 of the 10 runs did 150 iterations instead of 200 and are left out. In the 5 that did the same work, the power changed by −19.6 % and the run time by +9.7 %, so the energy changed by −11.8 %, the operational emissions by −11.8 %, the manufacture charged to the run by +9.7 %, and the footprint by +0.6 % (higher in every round).

Placement. In a second allocation on Aion, with its own baseline: Giving each process its own L3 cache changed the rating by +24.0 %, the run time by −18.2 % and the power by +4.2 %, so the footprint changed by −16.7 % (lower in every round) and the footprint per unit of work by −16.7 %. With the 2.0 GHz cap as well, the footprint changed by −21.7 % (lower in every round). The 2.0 GHz cap alone changed it by −4.7 % (lower in every round) in this allocation, against −3.9 % in the first.

The result. On IRIS, the change kept is the 2.6 GHz clock cap. Its 100 runs emitted 5.32 g per run (standard deviation 0.02 g), against 5.82 g for the 9 runs of the unchanged job on the same node, a change of −8.7 %. A run took 216.8 s instead of 214.1 s (+1.3 %), used 47.7 instead of 58.8 kJ (−18.9 %) and rated 12.4 GFLOP/s, the same to one decimal (−0.5 %); HPCG validated all of them, each with 100 iterations. The first of the 10 rounds on this node is left out of that comparison: its unchanged run took 245 s and rated 10.9 GFLOP/s, against 213 to 215 s and 12.4 to 12.5 GFLOP/s for the other 9, and the capped run of that round was slow too (225 s against 217 to 220 s); its clock and CPU use were those of the other runs, and nothing else the job recorded explains it. With that round, the change would be −9.8 %. In the rounds on this node, it emitted less than the unchanged job in 9 of 9 rounds, by 8.4 % on average. On another node, the 100 runs of Section III-C emitted 5.98 g, and this node's unchanged runs 2.5 % less: a difference between IRIS's nodes, which is why the comparison rests on the same node. On Aion, the change kept is one process per L3 cache with the 2.0 GHz cap. Its 100 runs emitted 2.51 g per run (standard deviation 0.009 g), against 3.13 g for the 100 runs of the unchanged job in Section III-C, a change of −20.0 %. A run took 111.0 s instead of 126.5 s (−12.2 %), used 27.8 instead of 40.1 kJ (−30.6 %) and rated 35.4 instead of 30.5 GFLOP/s (+16.3 %); HPCG validated all of them, each with 200 iterations. In the test rounds on one node, it emitted less than the unchanged job in 2 of 2 rounds, by 21.7 % on average.

D. What generalises

The method. Any job can be examined in four steps, using only the cluster's own levers. (1) Split the footprint of one run into its embodied and operational terms with the formula of Section III-C. Where manufacture is half of it or more, as on both clusters here, a change that saves energy but lengthens the run can cost more than it saves. (2) Choose the node type first: on the same job, moving from IRIS to Aion saved 48 %, far more than any setting tried on one node. (3) On that node type, try the levers that leave the code alone, in order of risk: compiler flags passed at build time, then the placement of the processes and a cap on the clock, both through the scheduler.

(4) Measure every candidate against the unchanged job in the same allocation, alternating them, and keep a change only if the program's own check still passes with at least the same work and if the total, not just the energy, goes down in every round; then measure the change kept with the full protocol.

The tool. Step (1) is carried out by a script, `jobprint`, which needs nothing but a job's identifier. It reads the job's record from the scheduler, or a saved copy of it when no cluster is at hand, and prints for each step the time and energy, the footprint split into node, site and manufacture, the largest of the three, and the reserved cores and memory the job left idle. Its repository, with a `README` and a saved record to try it on, is https://github.com/NexusSin/hpcg-ulhpc-carbon-footprint.git.

V. Carbon Payback

The cost of producing the optimisation has two parts. The first is the AI assistance: over the whole project, 3 888 124 output tokens, 430 g CO_2e at the course factor of 1.03 Wh per 1 000 output tokens and the grid factor of Section III-C; the session that set up and measured the optimisation accounts for 47 g of it. The second is the cluster time spent testing the configurations and measuring the kept one: the 120 runs on IRIS and 158 runs on Aion (plus 13 whose results were not used, in allocations stopped early) emitted 1 100 g with the factors of Section III-C. The cost is therefore between 1 100 g, counting that session alone, and 1 500 g, counting all the assistance.

The saving per run is the difference between the 100 runs after optimisation and the runs they are compared with in Section IV-C, 0.51 g on IRIS and 0.63 g on Aion. For an illustrative conversion to years, I counted other users' allocations with `hpcg` in the job or step name from October 2025 to October 2026. I assumed one comparable run per matching allocation and the same number of such allocations each year. On IRIS the estimated payback is 2 246 to 3 010 runs. The 7 matching allocations on IRIS would imply approximately 321 to 430 years under this assumption. On Aion the estimated payback is 1 819 to 2 438 runs. The 3 matching allocations on Aion would imply approximately 606 to 813 years under this assumption.

A. What this means

In answer to the research question: a run of HPCG emits 6.0 g CO_2e on IRIS and 3.1 g on Aion. The model attributes 36 % of the difference to site overhead. Of the settings a user controls without touching the code, the 2.6 GHz clock cap cuts a run by 8.7 % on IRIS and one process per L3 cache with the 2.0 GHz cap cuts a run by 20.0 % on Aion.

Because manufacture is half of the footprint or more, what counts is how long a run holds the node. A lower clock acts only on the operational part, 50 % of a run on IRIS and 42 % on Aion, and it lengthens the run, which charges more of the node's manufacture to it: on Aion the footprint fell at 2.0 GHz but rose at 1.5 GHz, where the runs that did the same work took longer still. Giving each process its own cache did the opposite: the same work finished sooner, so both terms fell, and the clock cap on top of it saved electricity while giving back only part of that time.

The optimisation cost 1 100 to 1 500 g. At the savings measured here, it would take about 1 819 to 3 010 comparable runs to recover that cost. Running this job on Aion with the original settings already reduced its footprint by 48 % relative to IRIS.

VI. Threats to Validity

The following checks examine how uncertainty in the measurements and model assumptions affects the comparison.

What the energy reading covers. On IRIS the counters cover the processor packages and the memory, on Aion the packages only. Nothing meters the power supply, fans, board, network card or local disk, which leaves the operational term too low on both clusters. Worse, the missing fans skew the comparison. Air-cooled IRIS nodes need them and liquid-cooled Aion nodes largely do without, so leaving them out favours IRIS. Reserving the node keeps other jobs out of the reading, but the cores the job leaves idle are charged to it. That weighs more on Aion, whose nodes have many more cores than the job uses.

Two machines, several differences. IRIS and Aion differ in processor, memory, node design and cooling at once. The formula estimates a facility overhead for each system. The comparison cannot isolate the effect of cooling technology or separate the effect of processor choice from memory configuration. Cooling acts inside the node as well: a hotter chip leaks more current, and that part of the cooling's effect is counted as processor energy, outside the PUE. Intel and AMD implement the counters differently, and the two readings may not be equally accurate. Aion reports no memory energy, while IRIS's figure includes 5.2 kJ of it per run: 0.25 g, or 9 % of the gap. On packages alone a run on IRIS would emit 5.7 g, 1.8 times Aion's, so the order of the clusters does not depend on it.

The declared PUE. Of all the threats, this one bears most directly on the research question. Since no measured PUE is published for either room, IRIS's value comes from European figures for rooms of its size and Aion's from its operator's design target. With a PUE of 1.43 or 1.83, a run on IRIS would emit 5.6 or 6.3 g. With 1.00, the target taken literally, or 1.55, the European average for rooms of Aion's size [10], a run on Aion would emit 3.0 or 3.7 g. Across these values the site overhead explains between 6 % and 46 % of the gap between the clusters: the part of the answer that concerns cooling rests on the PUE more than on any measurement. The European figure for small rooms comes partly from voluntary reports by more ambitious operators, which the assessment says may bias it low [10]; IRIS's true value may be higher. For that reason I also report the node alone, without any site overhead: 4.8 g on IRIS and 3.0 g on Aion.

The grid factor. Both clusters draw on the same grid. ILR's figure rests on what the suppliers declare: it counts 52 % of the supply as hydropower bought under contract, and only the carbon dioxide of generation [9]. With Ember's life-cycle

intensity of Luxembourg's own production, 123 g per kWh, a run would emit 6.4 g on IRIS and 3.3 g on Aion; with the EU-27 average of 209 g [15], 8.8 and 4.4 g. Ember counts upstream methane with a warming potential of 21, a value from an earlier IPCC report [15]. A higher factor raises the operational terms only: at the European average, the node's electricity, not its manufacture, would be the hotspot on both clusters. The ratio between the clusters barely moves: a run on IRIS emits 1.9 times as much as one on Aion at the declared factor, and 2.0 times at the European average.

Substitutes and completeness. Dell's sheet for the IRIS node states a standard deviation of ±6 410 kg on its 6 510 kg, assumes a four-year life and a configuration with 32 GB of memory where IRIS nodes hold 128 GB, and warns against its use in formal footprints [13]. Applied to the manufacture term, that standard deviation alone is ±2.9 g, about half of IRIS's total. The substitute for the Aion node models newer processors and four times its memory, so it probably overstates its manufacture. Nor are the figures equally complete. The embodied term covers manufacture, transport and end of life of the compute node. It excludes manufacture of switches, file servers and cooling equipment. Neither embodied source states which IPCC report its warming potentials follow. Most of these gaps make the total too low. The allocation adds its own uncertainty. Charging manufacture over the years in service so far charges a run the most it can carry: if IRIS's nodes stay in service for ten years, its manufacture term falls by an eighth. The allocated share is measured over the whole cluster, where many failed nodes of IRIS's older generation pull it down; the Skylake nodes alone are probably busier, which would lower their term further.

The functional unit. At a few minutes, the run is shorter and smaller than the HPCG rules require for an official result, and its figures hold for this configuration only. Running the same number of processes on both clusters keeps the job identical but leaves most of an Aion node idle, and a job sized to fill each node would compare the clusters differently. Each cluster was measured on one node; another node of the same model may run faster or slower, so the figures describe the node measured as much as the cluster.

The optimisation. On Aion, the 100 runs after optimisation ran on another node and day than the 100 runs before it, so their difference mixes the change with the difference between two nodes; Aion's nodes agree within a few per cent, and the test rounds, which compare the change with the unchanged job on one node, show how much of it is the change itself. On IRIS, whose nodes differ far more, the 100 capped runs are compared with 9 unchanged runs on the same node, fewer than 100, so that comparison is less precise than the one on Aion. The scheduler caps the cores' clock, but the memory controller and caches set their own, which a job can neither see nor set; a cap may act differently on another processor. `-march=native` ties the binary to the processor that built it. A single allocation can contain many runs, as the 100-run baseline jobs illustrate, and jobs with other names can also launch HPCG. The accounting records therefore leave the actual annual number of comparable runs unknown.

Burden shifting and the single criterion. A carbon-only result hides whatever it does not count. Liquid cooling may change how much water a site uses: at the European averages for the two size classes, 0.193 and 0.413 L per kWh of IT energy [10], a run consumes about 3.3 mL of water on IRIS and 4.6 mL on Aion, although direct liquid cooling can need little water, so Aion's figure may be too high. Replacing an air-cooled cluster with a liquid-cooled one means new hardware to make and old hardware to dispose of. The results cover climate change only, for one job on two clusters at one site, and support no comparison with other jobs.

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APPENDIX A
FIGURES AND TABLES

TABLE I
THE JOB AND THE TWO MACHINES IT RAN ON

	IRIS	Aion
Code	HPCG 3.1, https://github.com/hpcg-benchmark/hpcg [1], commit `114602d458d1034faa52b71e4c15aba9b3a17698` of 2023-09-23, reference kernels, unmodified	
Build	HPCG's `MPI_GCC_OMP` setup: `mpicxx -O3 -ffast-math -ftree-vectorize -fopenmp`; GCC 13.2.0 and Open MPI 4.1.6, built inside each allocation	
Input	`hpcg.dat`, 84 bytes, SHA-256 `8a18c3bde4485866bb24d3290da206003a4690112f2b01d6e77d9a3f7913376b`; line 3 `104 104 104` (grid points per process), line 4 `60` (target seconds of the timed phase)	
Job	28 MPI processes of one thread on one exclusive node, bound to cores; 100 runs back to back in one allocation per cluster, October 2026	
Partition	`batch`, Skylake nodes	`batch`
Node	Dell PowerEdge C6420	one of the three nodes of an Atos BullSequana X2410 blade
Processors	2 × Intel Xeon Gold 6132: 14 cores, 2.6 GHz, 140 W	2 × AMD EPYC 7H12: 64 cores, 2.6 GHz, 280 W
Memory	128 GB DDR4-2400, 4 of 6 channels per processor, peak 154 GB/s	256 GB DDR4-3200, all 8 channels per processor, peak 410 GB/s
Cooling, room	air, in-row units; 300 kW room	direct liquid, hot-water loop; 1 050 kW room
In service	Skylake nodes since December 2017	since late 2021
Energy read	RAPL counters: packages and memory	RAPL counters: packages only

Hardware, rooms and dates from the ULHPC documentation [14], [12], [11]; versions and checksum from the job logs. Peak memory bandwidth = channels × transfer rate × 8 bytes.

TABLE II
DECLARED FACTORS

Factor	Value	Source, basis
Grid intensity	108.42 gCO_2/kWh	ILR, supply to final customers, 2025, CO_2 only [9]
PUE, IRIS	1.63	EU, 100–500 kW class [10]
PUE, Aion	1.1	assumption: the operator's target for its liquid cooling, "very close to 1" [12]
Node manufacture, IRIS	1 256 kg CO_2e	Dell sheet, share outside use [13]
Node manufacture, Aion	1 200 kg CO_2e	modelled AMD node, substitute [7]
Allocated life, IRIS	8.8 y × 32.94 % = 2.90 y	ULHPC timeline; `sreport`
Allocated life, Aion	4.9 y × 54.37 % = 2.66 y	ULHPC timeline; `sreport`
AI assistance	1.03 Wh per 1 000 output tokens	course factor

Read 29 September and 23 October 2026. Comparators: 5 W LED at the grid factor, 291 g per km in a small petrol car, 28 kg per Luxembourg resident per day [15].

TABLE III
MEAN PER RUN OVER THE 100 RUNS ON EACH CLUSTER

Cluster	Runs	Time	Energy	Packages + memory	Power	Rating	Timed work	gCO_2e per run			
		(s)	(kJ)	(kJ)	(W)	(GFLOP/s)	(GFLOP)	node	site	manufacture	total
IRIS	100	217.3	61.0	55.8 + 5.2	281	12.3	1 173	1.8	1.2	3.0	6.0
Aion	100	126.5	40.1		317	30.5	2 345	1.2	0.12	1.8	3.1

Every value is the mean of the 100 runs on one node of each cluster. The footprint of a single run has a standard deviation of 0.03 g on IRIS and 0.06 g on Aion, and ranges from 5.86 to 6.06 g and from 3.08 to 3.25 g. Energy: RAPL counters read between runs. Timed work: the floating-point operations of HPCG's timed phase, 100 CG iterations per run on IRIS and 200 on Aion. Grams with the factors of Table II.

TABLE IV
MEAN FOOTPRINT OF A RUN, OVER 100 RUNS PER CLUSTER

gCO_2e per run	IRIS	Aion
Node electricity	1.8 (31 %)	1.2 (39 %)
Site overhead (PUE)	1.2 (19 %)	0.12 (4 %)
Node manufacture	3.0 (50 %)	1.8 (58 %)
Total	6.0	3.1
Standard deviation	0.03	0.06
Embodied : operational	1.0 : 1	1.4 : 1
Hours of a 5 W LED	11	5.8
Metres in a small petrol car	21	11
Share of a resident's day	0.021 %	0.011 %
At 123 g/kWh (Ember, LU)	6.4	3.3
At 209 g/kWh (EU-27)	8.8	4.4
embodied : operational	0.5 : 1	0.7 : 1
At PUE 1.43 / 1.83 (IRIS)	5.6 / 6.3	
At PUE 1.00 / 1.55 (Aion)		3.0 / 3.7
Node alone, no site overhead	4.8	3.0

100 runs on IRIS, 100 on Aion. Shares of the total in brackets. Operational = node electricity + site overhead.

TABLE V
OTHER MEASURED QUANTITIES: MEAN PER RUN OVER THE 100 RUNS

Quantity per run	IRIS	Aion
Cores held / used	28 / 28	128 / 28
Timed phase / rest of the run (s)	94 / 123	76 / 51
Peak power over 2 s (W)	291	322
Node CPU busy	99.6 %	21.8 %
Mean core clock (GHz)	3.3	2.8
Processor temperature, mean / hottest (°C)	74 / 81	61 / 65
Memory touched (GiB)	27.0	27.0
InfiniBand out / in (MB)	22.5 / 8.5	21.5 / 5.0
Ethernet out / in (MB)	0.16 / 0.19	0.15 / 0.13
Local SSD read / written (MB)	0.00 / 3.0	0.01 / 2.1
Idle node power, before and after (W)	94	179
Memory reserved (GiB)	112	219
Data between compute nodes	none, one node	none, one node
Saved HPCG output (kB/run)	7.08	7.15
Water, declared	3.3 mL	4.6 mL

Means of the 100 runs on each cluster, the node sampled every 2 s. Power: packages and memory on IRIS, packages only on Aion; idle power over 60 s before and after the runs. Water: litres per kWh of IT energy for each room's size class, 0.193 and 0.413 [10].

TABLE VI
UTILISATION OF THE TWO CLUSTERS OVER TWELVE MONTHS

Share of capacity	IRIS	Aion
Allocated, to jobs or reservations	32.94 %	54.37 %
Idle	29.06 %	35.86 %
Down	19.09 %	2.67 %
Planned	16.50 %	5.64 %
Planned down	2.41 %	1.47 %

`sreport cluster utilization`, 2025-09-29 to 2026-09-28, whole clusters. Many of IRIS's older nodes had failed, so its share probably understates the Skylake nodes' own.

TABLE VII
BEFORE AND AFTER OPTIMISATION: THE UNCHANGED JOB AND THE CHANGE KEPT, MEAN PER RUN

Cluster	Version	Runs	Time	Energy	Power	Rating	Iterations	gCO_2e per run			
			(s)	(kJ)	(W)	(GFLOP/s)		operational	manufacture	total	change
IRIS	unchanged, Section III-C	100	217.3	61.0	281	12.3	100	2.99	2.98	5.98	
	unchanged, same node	9	214.1	58.8	274	12.4	100	2.88	2.94	5.82	
	cap 2.6 GHz	100	216.8	47.7	220	12.4	100	2.34	2.98	5.32	−8.7 %
Aion	unchanged, Section III-C	100	126.5	40.1	317	30.5	200	1.33	1.81	3.13	
	spread, cap 2.0 GHz	100	111.0	27.8	251	35.4	200	0.922	1.58	2.51	−20.0 %

Same code, input, 28 processes, node held for the job alone, protocol and factors as Table III. Change: against the 100 runs of Section III-C on Aion, and against the unchanged runs on the same node on IRIS, whose nodes differ widely (Section IV-A). On IRIS, the first of the 10 rounds is left out (Section IV-C). Spread: one process per L3 cache. HPCG validated every run. gCO_2e to three digits so that the change shows; accurate to two (Section III-C).

TABLE VIII
THE ROUNDS THAT CHOSE THE CHANGE: EVERY CONFIGURATION AGAINST THE UNCHANGED JOB ON THE SAME NODE

Cluster	Configuration	Runs	Clock	Time	Energy	Power	Rating	gCO_2e per run			
			(GHz)	(s)	(kJ)	(W)	(GFLOP/s)	operational	manufacture	total	change
IRIS	unchanged	9	3.3	214.1	58.8	274	12.4	2.88	2.94	5.82	
	cap 2.6 GHz	9	2.6	217.9	47.7	219	12.3	2.34	2.99	5.34	−8.4 %
Aion (1)	unchanged	10	2.8	125.0	39.6	316	30.8	1.31	1.78	3.10	
	`-O2`	10	2.8	125.0	39.6	317	30.8	1.31	1.78	3.10	+0.1 %
	`-march=native`	10	2.8	124.9	39.5	316	30.9	1.31	1.78	3.09	−0.1 %
	cap 2.0 GHz	10	2.5	129.3	34.1	264	30.1	1.13	1.84	2.97	−3.9 %
	cap 1.5 GHz, 200 it.	5	2.4	137.2	34.9	254	28.5	1.16	1.96	3.11	+0.6 %
	cap 1.5 GHz, 150 it.	5	2.4	117.5	30.0	255	28.4	0.993	1.68	2.67	−13.8 %
Aion (2)	unchanged	2	2.7	125.3	41.1	328	30.8	1.36	1.79	3.15	
	cap 2.0 GHz	2	2.5	129.8	34.7	267	29.9	1.15	1.85	3.00	−4.7 %
	spread	2	2.7	102.4	35.0	342	38.2	1.16	1.46	2.62	−16.7 %
	spread, cap 2.0 GHz	2	2.0	109.0	27.4	252	36.1	0.908	1.56	2.46	−21.7 %

Means over the rounds of one allocation, on one node, each round running every configuration once in a rotating order. Unchanged: HPCG's own build (`-O3`) at the default clock and placement, as in Section III-B. Clock: mean over all the node's cores, sampled every 2 s; on Aion it averages the 28 cores that run HPCG with the 100 idle ones, so it stays above the cap, which the job recorded on the cores. Same code, input, 28 processes and factors as Table III; change relative to the unchanged job of the same allocation. (1), (2): Aion's two allocations; the second stopped after two rounds, once the result was clear. Spread: one process per L3 cache (Section IV-A); with the cap, it applied to all the node's cores. On IRIS, the cap was chosen beforehand, and its rounds give the unchanged job on the node of the 100 capped runs. On IRIS, the first of the 10 rounds is left out (Section IV-C). HPCG validated every run; "it.": a configuration whose runs did different numbers of iterations is split by that number (Section IV-C). gCO_2e to three digits, accurate to two.

APPENDIX B
AI USAGE JOURNAL

I used AI to help me read the technical documentation for IRIS and Aion and understand how the differences between their nodes affect this particular HPCG job. One question was what it meant to request 28 cores on each machine. On an IRIS node, the 28 processes occupy all the cores across its two Intel processors. On an Aion node, those same processes occupy 28 of the 128 cores across its two AMD processors. AI helped me interpret the specifications for memory channels and shared caches in the cluster documentation [14], [12]. These details matter for HPCG because its performance depends on how quickly each process can access the data it needs. I also used AI to adapt the SLURM submission script to the two clusters. This included understanding how the script requests an entire node, assigns one thread to each MPI process and binds processes to cores. The script gives each run its own directory and copies the same input file into it, so I could trace a benchmark report back to the run that produced it. Keeping the input and process count the same gave me a consistent basis for comparing the runs. Reserving the node for the job also meant that another user's workload would not contribute to the energy readings. Once the jobs had finished, I used AI to help me read the scheduler logs, the energy records and the reports produced by HPCG. Each source answered a different question about a run. The job record identified the node and allocation, the counter readings taken before and after the run gave the energy consumed by the components being monitored, and the benchmark report recorded the work completed and the performance achieved. Reading them together helped me relate a difference in elapsed time to the work done. It also helped me interpret a gap in the energy data. IRIS reports energy for both the processor packages and memory. Aion reports package energy only, so memory consumption on Aion is an unknown in the comparison. I also used AI to help extract information from the cited websites and reports and work out which published values applied to this job. For the electricity calculation, the Luxembourg regulator reports a factor for electricity supplied to customers, including imports [9]. That figure applies to the electricity used while the job runs. Dell's product footprint covers a server's whole life [13]. Using its full value would count the use phase twice, so the calculation uses the share outside that phase and allocates a fraction to the time the job occupied the node.